SET-B

Scholar Number:____________________ Name: ____________________

Maulana Azad National Institute of Technology Bhopal

Department of Electrical Engineering

MINI TEST EXAMINATION

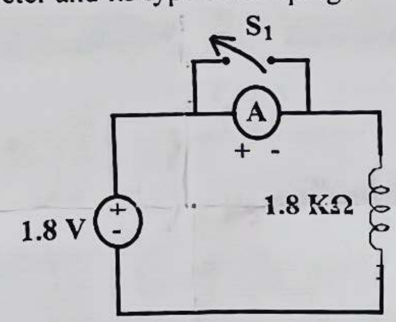
EE-214 MEASUREMENT AND INSTRUMENTATION

Subject Coordinator Names: Dr. Sushma Gupta & Dr. More Raju

Max Marks: 25 **Time: 1 Hour** **Date: 24/08/2023**

6 ⑤

1. Compare the digital and analog type of instrument. **[5 M]**
2. Explain indicating, integrating and recording type of instrument. **[5 M]**
3. Describe the primary and secondary transducer with suitable examples. **[5 M]**
4. A PMMC ammeter **A** in the figure shown below has a range of 0 to 3 mA. When switch S_1 is opened, the pointer of the ammeter swing to the 1 mA mark, returns and settles at 0.9 mA. Find the coil resistance of the meter and its type of damping. **[5 M]**



5. A PMMC voltmeter is connected across a series combination of DC voltage source $V_1 = 2$ V and voltage source $V_2(t) = 3\text{Sin}(314t)$ V. Find the meter reading. **[2 M]**
6. A PMMC instrument has a coil of dimensions 10 mm X 8 mm. The flux density in the air gap is 0.15 Wb/m^2. If the coil is wound for 100 turns, carrying a current of 5 mA, then calculate the deflecting torque. Calculate the deflection if the spring constant value is 0.2×10^{-6} N-m/degree. **[3 M]**